# 🛠️ Automated Customer Data Cleaning & Standardization Project

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## 📋 Project Objective

To automate the cleaning process of a "dirty" customer contact dataset. The raw data contains inconsistent formatting, duplicate entries, and missing values that hinder marketing analysis.

**Key Technical Goals:**

1. **Remove Duplicates:** Ensure unique customer records based on phone numbers.
2. **Standardize Formats:** Clean inconsistent phone number patterns.
3. **Data Imputation:** Handle `NaN` and null values for data integrity.

# 1. Data Acquisition & Inspection

Loading the raw dataset to identify inconsistencies and format errors.

In [1]:
```
import kagglehub

path = kagglehub.dataset_download("singhshivani0696/customer-call-list")

print("Path to dataset files:", path)
```

```
C:\Users\ASUS\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.9_qbz5n2kfra8p0\LocalCache\local-packages\Python39\site-packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
Path to dataset files: C:\Users\ASUS\.cache\kagglehub\datasets\singhshivani0696\customer-call-list\versions\1
```

In [2]:
```
import pandas as pd
import numpy as np
import os

file_path = os.path.join(path, "Customer Call List.xlsx")

df = pd.read_excel(file_path)
df
```

Out[2]:

| | CustomerID | First_Name | Last_Name | Phone_Number | Address | Paying Customer | Do_Not_Contact | Not_Useful_Column |
|---|---|---|---|---|---|---|---|---|
| **0** | 1001 | Frodo | Baggins | 123-545-5421 | 123 Shire Lane, Shire | Yes | No | True |
| **1** | 1002 | Abed | Nadir | 123/643/9775 | 93 West Main Street | No | Yes | False |
| **2** | 1003 | Walter | /White | 7066950392 | 298 Drugs Driveway | N | NaN | True |
| **3** | 1004 | Dwight | Schrute | 123-543-2345 | 980 Paper Avenue, Pennsylvania, 18503 | Yes | Y | True |
| **4** | 1005 | Jon | Snow | 876\|678\|3469 | 123 Dragons Road | Y | No | True |
| **5** | 1006 | Ron | Swanson | 304-762-2467 | 768 City Parkway | Yes | Yes | True |
| **6** | 1007 | Jeff | Winger | NaN | 1209 South Street | No | No | False |
| **7** | 1008 | Sherlock | Holmes | 876\|678\|3469 | 98 Clue Drive | N | No | False |
| **8** | 1009 | Gandalf | NaN | N/a | 123 Middle Earth | Yes | NaN | False |
| **9** | 1010 | Peter | Parker | 123-545-5421 | 25th Main Street, New York | Yes | No | True |
| **10** | 1011 | Samwise | Gamgee | NaN | 612 Shire Lane, Shire | Yes | No | True |
| **11** | 1012 | Harry | ...Potter | 7066950392 | 2394 Hogwarts Avenue | Y | NaN | True |
| **12** | 1013 | Don | Draper | 123-543-2345 | 2039 Main Street | Yes | N | False |
| **13** | 1014 | Leslie | Knope | 876\|678\|3469 | 343 City Parkway | Yes | No | False |
| **14** | 1015 | Toby | Flenderson_ | 304-762-2467 | 214 HR Avenue | N | No | False |
| **15** | 1016 | Ron | Weasley | 123-545-5421 | 2395 Hogwarts Avenue | No | N | False |
| **16** | 1017 | Michael | Scott | 123/643/9775 | 121 Paper Avenue, Pennsylvania | Yes | No | False |
| **17** | 1018 | Clark | Kent | 7066950392 | 3498 Super Lane | Y | NaN | True |
| **18** | 1019 | Creed | Braton | N/a | N/a | N/a | Yes | True |
| **19** | 1020 | Anakin | Skywalker | 876\|678\|3469 | 910 Tatooine Road, Tatooine | Yes | N | True |
| **20** | 1020 | Anakin | Skywalker | 876\|678\|3469 | 910 Tatooine Road, Tatooine | Yes | N | True |

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype 
---  ------             --------------  ----- 
 0   CustomerID         21 non-null     int64 
 1   First_Name         21 non-null     object
 2   Last_Name          20 non-null     object
 3   Phone_Number       19 non-null     object
 4   Address            21 non-null     object
 5   Paying Customer    21 non-null     object
 6   Do_Not_Contact     17 non-null     object
 7   Not_Useful_Column  21 non-null     bool  
dtypes: bool(1), int64(1), object(6)
memory usage: 1.3+ KB
```

In [4]:
```
df.duplicated().sum()
```

Out[4]: 1

## 2. Data Cleaning Process

Applying Python logic to transform the raw data into a usable format.

**Steps Taken:**

- **De-duplication:** Identified and removed duplicate rows based on the `Phone_Number` column to prevent redundant client contact.
- **Handling Null Values:** Imputed missing values in the `Do_Not_Contact` column (defaulting to "No") and replaced `NaN` / `N/a` with blank strings for cleaner visualization.
- **Normalization:** Resetting the index to maintain a structured sequence after row deletion.

In [5]:
```
# removing duplicates
df.drop_duplicates(inplace=True)
```

In [6]:
```
df.drop(columns='Not_Useful_Column', inplace=True)
```

In [7]:
```
# remove whitespace
df["First_Name"] = df["First_Name"].str.strip()
df["Last_Name"] = df["Last_Name"].str.strip("/._")
df[['First_Name','Last_Name']]
```

Out[7]:

| | First_Name | Last_Name |
|---|---|---|
| 0 | Frodo | Baggins |
| 1 | Abed | Nadir |
| 2 | Walter | White |
| 3 | Dwight | Schrute |
| 4 | Jon | Snow |
| 5 | Ron | Swanson |
| 6 | Jeff | Winger |
| 7 | Sherlock | Holmes |
| 8 | Gandalf | NaN |
| 9 | Peter | Parker |
| 10 | Samwise | Gamgee |
| 11 | Harry | Potter |
| 12 | Don | Draper |
| 13 | Leslie | Knope |
| 14 | Toby | Flenderson |
| 15 | Ron | Weasley |
| 16 | Michael | Scott |
| 17 | Clark | Kent |
| 18 | Creed | Braton |
| 19 | Anakin | Skywalker |

In [8]:
```
# remove character in phone number column
df['Phone_Number'] = df['Phone_Number'].astype(str)
df['Phone_Number'] = df['Phone_Number'].str.replace(r'\D', '', regex=True)
df['Phone_Number']
```

Out[8]:
```
0     1235455421
1     1236439775
2     7066950392
3     1235432345
4     8766783469
5     3047622467
6
7     8766783469
8
9     1235455421
10
11    7066950392
12    1235432345
13    8766783469
14    3047622467
15    1235455421
16    1236439775
17    7066950392
18
19    8766783469
Name: Phone_Number, dtype: object
```

In [9]:
```
# reformating phone number
df['Phone_Number'] = df['Phone_Number'].apply(lambda x: x[0:3]+'-'+x[3:7]+'-'+x[7:])
df['Phone_Number']
```

Out[9]:
```
0     123-5455-421
1     123-6439-775
2     706-6950-392
3     123-5432-345
4     876-6783-469
5     304-7622-467
6               --
7     876-6783-469
8               --
9     123-5455-421
10              --
11    706-6950-392
12    123-5432-345
13    876-6783-469
14    304-7622-467
15    123-5455-421
16    123-6439-775
17    706-6950-392
18              --
19    876-6783-469
Name: Phone_Number, dtype: object
```

In [10]:
```
df['Phone_Number'] = df['Phone_Number'].str.replace('--', '')
df['Phone_Number']
```

Out[10]:
```
0     123-5455-421
1     123-6439-775
2     706-6950-392
3     123-5432-345
4     876-6783-469
5     304-7622-467
6
7     876-6783-469
8
9     123-5455-421
10
11    706-6950-392
12    123-5432-345
13    876-6783-469
14    304-7622-467
15    123-5455-421
16    123-6439-775
17    706-6950-392
18
19    876-6783-469
Name: Phone_Number, dtype: object
```

In [11]:
```
df[["Street_Address", "State", "Zip_Code"]] = df["Address"].str.split(',', expand=True)
df.drop(columns='Address', inplace=True)
df
```

Out[11]:

| | CustomerID | First_Name | Last_Name | Phone_Number | Paying Customer | Do_Not_Contact | Street_Address | State | Zip_Code |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1001 | Frodo | Baggins | 123-5455-421 | Yes | No | 123 Shire Lane | Shire | None |
| **1** | 1002 | Abed | Nadir | 123-6439-775 | No | Yes | 93 West Main Street | None | None |
| **2** | 1003 | Walter | White | 706-6950-392 | N | NaN | 298 Drugs Driveway | None | None |
| **3** | 1004 | Dwight | Schrute | 123-5432-345 | Yes | Y | 980 Paper Avenue | Pennsylvania | 18503 |
| **4** | 1005 | Jon | Snow | 876-6783-469 | Y | No | 123 Dragons Road | None | None |
| **5** | 1006 | Ron | Swanson | 304-7622-467 | Yes | Yes | 768 City Parkway | None | None |
| **6** | 1007 | Jeff | Winger | | No | No | 1209 South Street | None | None |
| **7** | 1008 | Sherlock | Holmes | 876-6783-469 | N | No | 98 Clue Drive | None | None |
| **8** | 1009 | Gandalf | NaN | | Yes | NaN | 123 Middle Earth | None | None |
| **9** | 1010 | Peter | Parker | 123-5455-421 | Yes | No | 25th Main Street | New York | None |
| **10** | 1011 | Samwise | Gamgee | | Yes | No | 612 Shire Lane | Shire | None |
| **11** | 1012 | Harry | Potter | 706-6950-392 | Y | NaN | 2394 Hogwarts Avenue | None | None |
| **12** | 1013 | Don | Draper | 123-5432-345 | Yes | N | 2039 Main Street | None | None |
| **13** | 1014 | Leslie | Knope | 876-6783-469 | Yes | No | 343 City Parkway | None | None |
| **14** | 1015 | Toby | Flenderson | 304-7622-467 | N | No | 214 HR Avenue | None | None |
| **15** | 1016 | Ron | Weasley | 123-5455-421 | No | N | 2395 Hogwarts Avenue | None | None |
| **16** | 1017 | Michael | Scott | 123-6439-775 | Yes | No | 121 Paper Avenue | Pennsylvania | None |
| **17** | 1018 | Clark | Kent | 706-6950-392 | Y | NaN | 3498 Super Lane | None | None |
| **18** | 1019 | Creed | Braton | | N/a | Yes | N/a | None | None |
| **19** | 1020 | Anakin | Skywalker | 876-6783-469 | Yes | N | 910 Tatooine Road | Tatooine | None |

In [12]:

```
# Reordering the columns
ls = list(df.columns.values)
df = df[ls[0:4] + ls[6:]+ls[4:6]]
df
```

Out[12]:

| | CustomerID | First_Name | Last_Name | Phone_Number | Street_Address | State | Zip_Code | Paying Customer | Do_Not_Contact |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1001 | Frodo | Baggins | 123-5455-421 | 123 Shire Lane | Shire | None | Yes | No |
| **1** | 1002 | Abed | Nadir | 123-6439-775 | 93 West Main Street | None | None | No | Yes |
| **2** | 1003 | Walter | White | 706-6950-392 | 298 Drugs Driveway | None | None | N | NaN |
| **3** | 1004 | Dwight | Schrute | 123-5432-345 | 980 Paper Avenue | Pennsylvania | 18503 | Yes | Y |
| **4** | 1005 | Jon | Snow | 876-6783-469 | 123 Dragons Road | None | None | Y | No |
| **5** | 1006 | Ron | Swanson | 304-7622-467 | 768 City Parkway | None | None | Yes | Yes |
| **6** | 1007 | Jeff | Winger | | 1209 South Street | None | None | No | No |
| **7** | 1008 | Sherlock | Holmes | 876-6783-469 | 98 Clue Drive | None | None | N | No |
| **8** | 1009 | Gandalf | NaN | | 123 Middle Earth | None | None | Yes | NaN |
| **9** | 1010 | Peter | Parker | 123-5455-421 | 25th Main Street | New York | None | Yes | No |
| **10** | 1011 | Samwise | Gamgee | | 612 Shire Lane | Shire | None | Yes | No |
| **11** | 1012 | Harry | Potter | 706-6950-392 | 2394 Hogwarts Avenue | None | None | Y | NaN |
| **12** | 1013 | Don | Draper | 123-5432-345 | 2039 Main Street | None | None | Yes | N |
| **13** | 1014 | Leslie | Knope | 876-6783-469 | 343 City Parkway | None | None | Yes | No |
| **14** | 1015 | Toby | Flenderson | 304-7622-467 | 214 HR Avenue | None | None | N | No |
| **15** | 1016 | Ron | Weasley | 123-5455-421 | 2395 Hogwarts Avenue | None | None | No | N |
| **16** | 1017 | Michael | Scott | 123-6439-775 | 121 Paper Avenue | Pennsylvania | None | Yes | No |
| **17** | 1018 | Clark | Kent | 706-6950-392 | 3498 Super Lane | None | None | Y | NaN |
| **18** | 1019 | Creed | Braton | | N/a | None | None | N/a | Yes |
| **19** | 1020 | Anakin | Skywalker | 876-6783-469 | 910 Tatooine Road | Tatooine | None | Yes | N |

In [ ]:
```
df['Paying Customer'] = df['Paying Customer'].replace({'Y': 'Yes', 'N': 'No'})
df['Do_Not_Contact'] = df['Do_Not_Contact'].replace({'Y': 'Yes', 'N': 'No'})
df
```

Out[ ]:

| | CustomerID | First_Name | Last_Name | Phone_Number | Street_Address | State | Zip_Code | Paying Customer | Do_Not_Contact |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1001 | Frodo | Baggins | 123-5455-421 | 123 Shire Lane | Shire | None | Yes | No |
| **1** | 1002 | Abed | Nadir | 123-6439-775 | 93 West Main Street | None | None | No | Yes |
| **2** | 1003 | Walter | White | 706-6950-392 | 298 Drugs Driveway | None | None | No | NaN |
| **3** | 1004 | Dwight | Schrute | 123-5432-345 | 980 Paper Avenue | Pennsylvania | 18503 | Yes | Yes |
| **4** | 1005 | Jon | Snow | 876-6783-469 | 123 Dragons Road | None | None | Yes | No |
| **5** | 1006 | Ron | Swanson | 304-7622-467 | 768 City Parkway | None | None | Yes | Yes |
| **6** | 1007 | Jeff | Winger | | 1209 South Street | None | None | No | No |
| **7** | 1008 | Sherlock | Holmes | 876-6783-469 | 98 Clue Drive | None | None | No | No |
| **8** | 1009 | Gandalf | NaN | | 123 Middle Earth | None | None | Yes | NaN |
| **9** | 1010 | Peter | Parker | 123-5455-421 | 25th Main Street | New York | None | Yes | No |

In [14]:
```
# drop Do_not_contact (Yes) and phone_number (missing value)
df = df[df['Do_Not_Contact'] != 'Yes']
df = df[df['Phone_Number'] != '']
df = df.reset_index(drop=True)
df
```

Out[14]:

| | CustomerID | First_Name | Last_Name | Phone_Number | Street_Address | State | Zip_Code | Paying Customer | Do_Not_Contact |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1001 | Frodo | Baggins | 123-5455-421 | 123 Shire Lane | Shire | None | Yes | No |
| **1** | 1003 | Walter | White | 706-6950-392 | 298 Drugs Driveway | None | None | No | NaN |
| **2** | 1005 | Jon | Snow | 876-6783-469 | 123 Dragons Road | None | None | Yes | No |
| **3** | 1008 | Sherlock | Holmes | 876-6783-469 | 98 Clue Drive | None | None | No | No |
| **4** | 1010 | Peter | Parker | 123-5455-421 | 25th Main Street | New York | None | Yes | No |
| **5** | 1012 | Harry | Potter | 706-6950-392 | 2394 Hogwarts Avenue | None | None | Yes | NaN |
| **6** | 1013 | Don | Draper | 123-5432-345 | 2039 Main Street | None | None | Yes | No |
| **7** | 1014 | Leslie | Knope | 876-6783-469 | 343 City Parkway | None | None | Yes | No |
| **8** | 1015 | Toby | Flenderson | 304-7622-467 | 214 HR Avenue | None | None | No | No |
| **9** | 1016 | Ron | Weasley | 123-5455-421 | 2395 Hogwarts Avenue | None | None | No | No |
| **10** | 1017 | Michael | Scott | 123-6439-775 | 121 Paper Avenue | Pennsylvania | None | Yes | No |
| **11** | 1018 | Clark | Kent | 706-6950-392 | 3498 Super Lane | None | None | Yes | NaN |
| **12** | 1020 | Anakin | Skywalker | 876-6783-469 | 910 Tatooine Road | Tatooine | None | Yes | No |

In [15]:
```
df["Do_Not_Contact"] = df["Do_Not_Contact"].replace('', 'No')
df = df.reset_index(drop=True)
df
```

Out[15]:

| | CustomerID | First_Name | Last_Name | Phone_Number | Street_Address | State | Zip_Code | Paying Customer | Do_Not_Contact |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1001 | Frodo | Baggins | 123-5455-421 | 123 Shire Lane | Shire | None | Yes | No |
| **1** | 1003 | Walter | White | 706-6950-392 | 298 Drugs Driveway | None | None | No | NaN |
| **2** | 1005 | Jon | Snow | 876-6783-469 | 123 Dragons Road | None | None | Yes | No |
| **3** | 1008 | Sherlock | Holmes | 876-6783-469 | 98 Clue Drive | None | None | No | No |
| **4** | 1010 | Peter | Parker | 123-5455-421 | 25th Main Street | New York | None | Yes | No |
| **5** | 1012 | Harry | Potter | 706-6950-392 | 2394 Hogwarts Avenue | None | None | Yes | NaN |
| **6** | 1013 | Don | Draper | 123-5432-345 | 2039 Main Street | None | None | Yes | No |
| **7** | 1014 | Leslie | Knope | 876-6783-469 | 343 City Parkway | None | None | Yes | No |
| **8** | 1015 | Toby | Flenderson | 304-7622-467 | 214 HR Avenue | None | None | No | No |
| **9** | 1016 | Ron | Weasley | 123-5455-421 | 2395 Hogwarts Avenue | None | None | No | No |
| **10** | 1017 | Michael | Scott | 123-6439-775 | 121 Paper Avenue | Pennsylvania | None | Yes | No |
| **11** | 1018 | Clark | Kent | 706-6950-392 | 3498 Super Lane | None | None | Yes | NaN |
| **12** | 1020 | Anakin | Skywalker | 876-6783-469 | 910 Tatooine Road | Tatooine | None | Yes | No |

In [16]:

```
df["Do_Not_Contact"] = df["Do_Not_Contact"].fillna("No")
df = df.drop_duplicates(subset="Phone_Number", keep="first")
df = df.reset_index(drop=True)
df
```

Out[16]:

| | CustomerID | First_Name | Last_Name | Phone_Number | Street_Address | State | Zip_Code | Paying Customer | Do_Not_Contact |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1001 | Frodo | Baggins | 123-5455-421 | 123 Shire Lane | Shire | None | Yes | No |
| **1** | 1003 | Walter | White | 706-6950-392 | 298 Drugs Driveway | None | None | No | No |
| **2** | 1005 | Jon | Snow | 876-6783-469 | 123 Dragons Road | None | None | Yes | No |
| **3** | 1013 | Don | Draper | 123-5432-345 | 2039 Main Street | None | None | Yes | No |
| **4** | 1015 | Toby | Flenderson | 304-7622-467 | 214 HR Avenue | None | None | No | No |
| **5** | 1017 | Michael | Scott | 123-6439-775 | 121 Paper Avenue | Pennsylvania | None | Yes | No |

## 3. Final Output & Export

The data is now clean, consistent, and duplicate-free. The final dataset is exported to CSV/Excel for further analysis or CRM integration.

In [17]:

```
df.to_excel("Customer_Call_List_Cleaned.xlsx", index=False)
df
```

Out[17]:

| | CustomerID | First_Name | Last_Name | Phone_Number | Street_Address | State | Zip_Code | Paying Customer | Do_Not_Contact |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 1001 | Frodo | Baggins | 123-5455-421 | 123 Shire Lane | Shire | None | Yes | No |
| **1** | 1003 | Walter | White | 706-6950-392 | 298 Drugs Driveway | None | None | No | No |
| **2** | 1005 | Jon | Snow | 876-6783-469 | 123 Dragons Road | None | None | Yes | No |
| **3** | 1013 | Don | Draper | 123-5432-345 | 2039 Main Street | None | None | Yes | No |
| **4** | 1015 | Toby | Flenderson | 304-7622-467 | 214 HR Avenue | None | None | No | No |
| **5** | 1017 | Michael | Scott | 123-6439-775 | 121 Paper Avenue | Pennsylvania | None | Yes | No |